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Emily Fitzwilliam[1,2], Nikolai Steventon-Lorenzen[1,2], David Opar[1,2], Anthony G. Schache[3], and Nirav Maniar[1,2]

[1]School of Behavioural and Health Sciences, Australian Catholic University, Victoria, AUSTRALIA; [2]Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Faculty of Health Sciences, Australian Catholic University, Victoria, AUSTRALIA; [3]La Trobe Sport and Exercise Medicine Research Centre (LASEM), La Trobe University, Melbourne, Victoria, AUSTRALIA

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Emily Fitzwilliam[1,2], Nikolai Steventon-Lorenzen[1,2], David Opar[1,2], Anthony G. Schache[3], and Nirav Maniar[1,2]

[1]School of Behavioural and Health Sciences, Australian Catholic University, Victoria, AUSTRALIA; [2]Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Faculty of Health Sciences, Australian Catholic University, Victoria, AUSTRALIA; [3]La Trobe Sport and Exercise Medicine Research Centre (LASEM), La Trobe University, Melbourne, Victoria, AUSTRALIA

Address for correspondence: Dr Nirav Maniar, School of Behavioural and Health Sciences, Australian Catholic University, 115 Victoria Parade, Fitzroy, Melbourne, Australia 3065; E-mail: Nirav.Maniar@acu.edu.au

Conflict of Interest and Funding Source:

This study was partly funded from the Faculty of Health Sciences Funding Support for Honours and Masters by Coursework with Minor Thesis scheme, provided by the Australian Catholic University. A/Prof David Opar is listed as a co-inventor on a patent, filed by the Queensland University of Technology (QUT), for a field-testing device of eccentric hamstring strength, which is now known commercially as the NordBord. A/Prof Opar has received revenue

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distributions from QUT based on revenue that QUT has generated through the commercialisation of his intellectual property. A/Prof Opar is a minority shareholder in Vald Performance Pty Ltd, the company responsible for commercialization of the NordBord and the SmartSpeed timing gates, the latter which were utilised in the current manuscript. A/Prof Opar has received research funding from Vald Performance, for work unrelated to the current manuscript. A/Prof Opar was previously the Chair of the Vald Performance Research Committee, a role that was unpaid.

A/Prof Opar has family members who are minor shareholders and/or employees of Vald Performance.

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ABSTRACT

Introduction: Maximal acceleration and deceleration tasks are frequently required in team sports, often occurring rapidly in response to external stimuli. Accelerating and decelerating can be associated with lower limb injuries, thus knowledge of joint mechanics during these tasks can improve the understanding of both human high performance and injury mechanisms. The current study investigated the fundamental differences in lower limb joint mechanics when accelerating and decelerating by directly comparing the hip, knee and ankle joint moments and work done between the two tasks.

Methods: Twenty participants performed maximal effort acceleration and deceleration trials, with three-dimensional marker trajectories and ground reaction forces collected simultaneously. Experimental data was combined with inverse dynamics analysis to compute joint moments and work.

Results: Net joint work for all lower limb joints was positive during acceleration, and negative during deceleration. This occurred due to significantly greater positive work production from the ankle and hip during acceleration, and significantly greater negative work production from all joints during deceleration. The largest contributions to positive work during acceleration came from the ankle, followed by the hip and knee joints; whilst the largest contributions to negative work during deceleration came from the knee and hip joints, followed by the ankle. Peak joint moments were significantly greater when decelerating compared to accelerating, except for the peak ankle plantarflexion and hip flexion moments which were significantly greater when

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accelerating.

Conclusions: Our findings may help to guide training interventions which aim to enhance the performance of acceleration and deceleration tasks, whilst also mitigating the associated injury risk.

Key Words: JOINT WORK, JOINT POWER, JOINT MOMENTS, GAIT BIOMECHANICS, INVERSE DYNAMICS

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INTRODUCTION

Team sports frequently require rapid changes in running velocity in response to unanticipated external demands of play (1). These velocity changes may be increases (i.e., acceleration) or decreases (i.e., deceleration), both of which are crucial for sports performance and thus match outcome (1). For example, for all player positions in Australian Rules football, the total amount of acceleration and deceleration bouts during a typical match can range from 82-125 (2). Acceleration and deceleration are also common mechanisms for injury, particularly to the knee joint (3), hamstring muscles (4) and calf muscles (5), probably due to the high forces encountered during these tasks (4-7). These injuries can result in further long-term health complications (e.g., osteoarthritis in the case of a knee injury (8)), reduced sports performance (9), and may also place substantial financial burdens on both the individual (10) and sporting club (11). Therefore, a thorough understanding of human acceleration and deceleration mechanics has direct implications for both improving sports performance and minimising injury risk.

The purpose of accelerating and decelerating is to change centre of mass (COM) velocity (12). These velocity changes are produced by skeletal muscular contractions, which subsequently produce internal joint moments (13). Joint moments are rotational forces around a joint that cause movement of the limbs which, in the case of the weight-bearing lower limb, results in the foot exerting a force onto the ground during the stance phase of gait (i.e., where the foot is in contact with the ground). Owing to Newton's third Law, an equal and opposite force to that exerted by the foot is then produced (i.e., a ground reaction force (GRF)). This GRF can change COM velocity depending on its magnitude and direction, as per Newton's second law (i.e., force

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is the product of mass and acceleration). An anteriorly directed net GRF causes COM forward acceleration, whereas a posteriorly directed net GRF accelerates the COM backwards (or decelerates the forward momentum of the body). The magnitude and direction of the net GRF acting on a body depends on the relative joint moment contributions and orientation of all the segments in the body system (14). Therefore, elucidating the joint mechanics and GRFs when accelerating and decelerating can provide direct insight into the performance of such tasks.

Previous research has found that the hip extension and ankle plantarflexion moments accelerate the COM anteriorly (15) and the angular impulse of both moments are associated with forward acceleration magnitude (16). Furthermore, the largest contributors to net positive joint work when accelerating are the ankle and hip (12). In comparison, deceleration requires substantial negative joint work to reduce the COM forward velocity, with the knee and ankle being the largest contributors to net negative joint work (12). Despite providing some important insights, the forementioned studies have some limitations. Firstly, these studies have low sample sizes (range = 7-10) (12, 15, 16), limiting the statistical power and generalisability of the research (17). Secondly, none of the studies directly compared the joint mechanics between accelerative and decelerative running (12, 15, 16, 18), even if both tasks were included in the study (12). Lastly, none of the studies tested maximal effort and unanticipated deceleration, as would be expected in team sport settings (1).

Therefore, this study aimed to directly compare lower limb joint mechanics during maximal acceleration and deceleration. Based on the findings of previous work (12, 16), it was hypothesised that 1) the net joint work produced by the lower limbs would be positive during

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acceleration and negative during deceleration, and 2) the ankle and hip would produce the most positive joint work during acceleration, whereas the knee and ankle would produce the most negative joint work during deceleration.

METHODS

Sample

A total of twenty participants (10 males and 10 females) were recruited for the current study through convenience and volunteer sampling methods (age: 25 ± 4 years; height: 174 ± 8 cm; mass: 70 ± 10 kg). The calculation for sample size was performed using the "pwr" package in the R programming language (19). Although no previous studies have compared joint mechanics during maximal effort acceleration and deceleration, one study reported joint work during gradual acceleration and deceleration (albeit without direct comparisons between tasks) (12). Of the data reported that were related to our hypotheses (i.e., ankle and hip positive joint work; knee and ankle negative joint work) a range of anticipated Cohen's d effect sizes were computed (1.04 - 2.47). Based on this, we adopted conservative input parameters ($d = 1.0$, $\alpha = 0.01$ and power = 0.90) for our sample size calculation, informing a target of twenty participants (see Supplemental Figure 1, Supplemental Digital Content, http://links.lww.com/MSS/C997). Conservative inputs were chosen because 1) our analysis included more variables than just joint work (e.g., peak joint moments) which were not reported previously (12); and 2) the study used as a basis for sample size calculations (12) had a small sample size (ten participants), thus their effect size estimates may be inflated due to lowered statistical power (20). All participants were required to be recreationally active and free from any injury that could affect their ability to perform the experimental protocol, to support generalisability of our findings and minimise

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potential confounders. To be considered recreationally active, participants were required to engage in at least 150 minutes of moderate physical activity per week, as determined using Exercise and Sport Science Australia pre-screening questionnaire. Each participant provided written, informed consent prior to study commencement. The Australian Catholic University Human Research Ethics Committee granted ethical approval to conduct the study (approval number: 2019-351H).

Experimental set up

The experiment was conducted in the biomechanics laboratory at the Melbourne campus of the Australian Catholic University. The laboratory is equipped with two floor-embedded force plates (AMTI, Massachusetts, USA) to record GRF data at a sampling rate of 1000Hz. A 10-camera three-dimensional (3D) motion analysis system (Vicon, Oxford Metrics., Oxford, UK) was used to measure the trajectories of skin-mounted reflective markers (sampled at 200Hz) to track the participants' motion throughout the dynamic trials. Two sets of timing gates (SmartSpeed, Vald Performance, Newstead, Australia) were positioned on either side of the force plates at a set distance of 3m apart (as determined via pilot testing).

Experimental protocol

Participants were asked to wear exercise appropriate clothing and footwear to enable completion of the experimental protocol. Participants were allowed to choose their preferred/usual running footwear to avoid potentially disturbing their natural movement strategy. Prior to commencing the experimental trials, 47 individual skin-mounted reflective markers (14mm diameter) and four marker clusters (each with 4 markers) were placed on key anatomical

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locations of the body, similar to previous work (21) (Figure 1b). Initially, participants completed a static trial where they stood completely still on one of the force plates in the anatomical position. Upon finishing the static trials, four reflective markers (medial malleoli and medial femoral epicondyles) were removed prior to starting the dynamic trials so as to not impede the participants' movement.

The participants then completed a standardised warm-up before commencing the maximal effort acceleration trials. Participants were positioned approximately 2-3 steps behind the first force plate and were required to accelerate with maximal effort through both sets of timing gates from a two-point, stationary start. Each participant's starting position was then manually adjusted by the investigator until their right foot naturally landed within the boundaries of the force plate on the second or third step of the acceleration. During these trials, timing gates were only used to monitor split times as a marker of average running velocity. Trials were repeated, with adequate recovery to minimise the effects of fatigue, until three successful trials were recorded. To be deemed a successful trial, the participant needed to have the entirety of their right foot land within the perimeter of a force plate. Next, participants completed the deceleration trials. For this task, participants began each trial like the acceleration trials, however immediately upon breaking the first set of timing gates, the second gate was then triggered to flash either green (indicating to continue accelerating) or red (indicating to immediately stop, i.e., decelerate). The colour flashed by the second set of timing gates was random and unknown to both the researcher and the participant. This experimental setup enabled the capture of maximal effort and unanticipated deceleration of the same step (second or third step for the right lower limb) captured in the initial acceleration trials for each participant. For any trials where the

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timing gates flashed green, investigators monitored split times recorded by the timing gates to ensure it approximated the initial acceleration trials. If trials were slower than the initially recorded acceleration trials, feedback was given to participants to encourage maximal effort. Trials were repeated until three successful deceleration trials were completed. Owing to different stride patterns during deceleration, the starting position of the participant (and the timing gates) was again adjusted by the investigator where necessary to allow each participant's right foot to fall naturally within the boundary of the force plate.

Data analysis

A total of 120 trials (three acceleration and three deceleration trials per participant) were successfully recorded for subsequent analysis. Prior to biomechanical analysis, both marker trajectory and GRF data from all trials were low pass filtered through a zero-lag 4th order Butterworth filter with a cut-off frequency of 15Hz (22). Biomechanical analysis was then completed using the OpenSim (Version 4.2) application (23, 24). For each participant, a generic 31 degrees of freedom (DOF) full-body musculoskeletal model (25, 26) was scaled to match the specific anthropometry of each participant, as determined during the static trial. The model included a single rigid head-trunk segment, which articulated with the pelvis via a 3-DOF ball and socket joint. The hip joints were modelled as 3-DOF ball and socket joints. The knee joints were represented as 1-DOF hinge joints, where both rotational (valgus/varus and internal/external rotation) and translational (antero-posterior, superior-inferior and mediolateral) motions were limited to change as a function of the knee flexion angle (27). The ankle (i.e., talocrural) and subtalar joints were modelled as two non-intersecting, 1-DOF pin joints. Although not reported in this study, the upper limbs were also modelled, including 3-DOF ball

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and socket shoulder joints, and separate 1-DOF elbow and radioulnar joints. For dynamic trials, inverse kinematics was used to calculate the joint angles during each trial. Subsequent inverse dynamics was used to determine the net internal lower limb joint moments (N·m). Joint power (W) was then calculated as follows (27):

$$P_j = M_j \cdot \omega_j \qquad [1]$$

where P_j is the net joint power, M_j is the net joint moment (computed from inverse dynamics) and ω_j is the joint angular velocity (computed as the first derivative of joint angular position with respect to time). The net work (J) completed by a single joint was then determined as follows (27):

$$W_j = \int_{t_0}^{t_1} P_j \, dt \qquad [2]$$

where W_j is the net joint work, P_j is the net joint power, t_0 is the beginning of the time period of interest and t_1 is the completion of the time period of interest.

Outcome Variables

The primary outcome measures of interest were mechanical work (i.e., net joint work, and total positive and negative work) and the peak joint moments at the hip, knee and ankle (talocrural) joints during the stance phase of acceleration and deceleration (16, 18). Note that since joint power/work is scalar, hip joint power was computed as the sum of its three orthogonal components, whilst all other joints were modelled as 1-DOF joints and hence did not require this

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step. Analysis of peak joint moments was restricted to the sagittal plane (i.e., flexion/extension) for our main analysis, owing to its relevance to acceleration/deceleration (12, 15, 16, 18). Non-sagittal lower limb mechanics (i.e., at the hip and subtalar joints) are provided in Supplemental Figure 2 (Supplemental Digital Content, http://links.lww.com/MSS/C997) for the interested reader. Joint kinetics were normalised to body mass, such that joint moments, power and work were expressed as $N \cdot m \cdot kg^{-1}$, $W \cdot kg^{-1}$ and $J \cdot kg^{-1}$, respectively. The GRF data was normalised to bodyweight (BW) for visualisation and was also used to determine the net COM acceleration magnitude by averaging the anteroposterior component across stance (and dividing by body mass) (16). Calculating the net acceleration magnitude enabled investigators to verify if the task objectives (i.e., acceleration or deceleration) were achieved. This verification process identified two deceleration trials (from two unique participants) that were not truly decelerative. These trials were subsequently excluded, leaving a total of 118 trials included for all subsequently described statistical analyses. To inform the reader of the execution of each task, the forward COM velocity at foot contact was also computed from our kinematic model.

Statistical analysis

R software (19) was used to complete the statistical analysis. Linear mixed models were used to determine if the peak joint moments, net joint work, net positive joint work and net negative joint work at the hip, knee and ankle joints were significantly different for acceleration versus deceleration (fixed effect), whilst accounting for participant identification (random effect). For all analyses, a conservative statistical significance was set at $p < .01$ (two-tailed), owing to the number of outcome variables investigated. The test assumptions were assessed via visual inspection of residual plots and QQ-plots. Data on the primary outcome measures of

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interest (joint moments and work) were reported as the marginal mean and 95% confidence interval of acceleration and deceleration conditions, as well as the z score estimate and 95% confidence interval for differences between acceleration and deceleration conditions. Other descriptive data (e.g., net COM acceleration magnitude) were reported as mean ± 1SD.

RESULTS

The mean ± 1SD forward velocity of the COM at foot strike for the acceleration and deceleration tasks was 3.91 ± 0.43 and 3.70 ± 0.52 $m \cdot s^{-1}$, respectively. The mean ± 1SD magnitude of the COM anteroposterior acceleration during stance was 3.23 ± 0.87 $m \cdot s^{-2}$ for the acceleration trials (i.e., positive=forwards/anterior), whereas it was -3.67 ± 1.44 $m \cdot s^{-2}$ for the deceleration trials (i.e., negative=backwards/posterior). The group mean kinematics for acceleration and deceleration, as well as the GRFs, are visualised in Figure 2. The joint angle, moment and power profiles are shown in Figure 3 whereas Figure 4 depicts net joint work in relation to the net COM acceleration magnitude.

Hip

During both acceleration and deceleration, there was a net hip extension moment until 71% (deceleration) and 72% (acceleration) of stance, with a small hip flexion moment for the remainder of stance (Figure 3d). However, there was a large difference ($z = 1.50$, 95% CI = 1.31 to 1.68, $p < 0.001$) in the magnitude of the hip extension moment peak during deceleration (mean = -4.97, 95% CI = -5.46 to -4.48 $N \cdot m \cdot kg^{-1}$) compared to that during acceleration (mean = -1.88, 95% CI = -2.38 to -1.39 $N \cdot m \cdot kg^{-1}$) (Table 1). Across the hip extension moment period, power was mostly positive (generation) in acceleration and negative (absorption) during deceleration

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(Figure 3g). This resulted in a large difference in the net joint work between conditions ($z = 1.46$, 95% CI = 1.24 to 1.68, $p < 0.001$), where the net joint work across stance was positive when accelerating (mean = 0.48 $J \cdot kg^{-1}$, 95% CI = 0.29 to 0.66 $J \cdot kg^{-1}$), and negative when decelerating (mean = -0.73 $J \cdot kg^{-1}$; 95% CI =-0.91 to -0.55 $J \cdot kg^{-1}$) (Table 1).

Knee

There was a net extension joint moment at the knee for both conditions, except for a flexion moment during early stance (2-13%) for deceleration and during late stance (85-98%) for acceleration (Figure 3e). The difference in the knee extension moment peak was large ($z = 1.07$, 95% CI = 0.85 to 1.29, $p < 0.001$) between acceleration (mean = -1.66, 95% CI = -1.85 to -1.47 $N \cdot m \cdot kg^{-1}$) and deceleration (mean = -2.30, 95% CI = -2.49 to -2.10 $N \cdot m \cdot kg^{-1}$) (Table 1). When accelerating, power fluctuated between periods of generation (0-13%, 37-85% and 98-100% of stance), and absorption (14-36, and 86-97% of stance) (Figure 3h). When decelerating, power was generated within the first 13% of stance, then absorbed for the remainder of stance (Figure 3j). Overall, this led to a large difference in net joint work ($z = 1.50$, 95% CI = 1.28 to 1.73, $p < 0.001$) (Table 1) where positive joint work was done when accelerating (mean = 0.37 $J \cdot kg^{-1}$, 95% CI = 0.24 to 0.50 $J \cdot kg^{-1}$), whereas negative joint work was done when decelerating (mean = -0.65 $J \cdot kg^{-1}$, 95% CI = -0.78 to -0.52 $J \cdot kg^{-1}$) (Table 1).

Ankle

Apart from a dorsiflexion moment from foot strike until 8% of stance when decelerating, both conditions displayed a net plantarflexion moment at the ankle throughout stance (Figure 3f). There was a large difference ($z = -1.61$, 95% CI = -1.76 to -1.46, $p < 0.001$) in the magnitude of

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the peak plantarflexion moment between acceleration (mean = -2.86 $N \cdot m \cdot kg^{-1}$, 95% CI = -3.01 to -2.70 $N \cdot m \cdot kg^{-1}$) compared to deceleration (mean = -1.76 $N \cdot m \cdot kg^{-1}$, 95% CI = -1.91 to -1.60 $N \cdot m \cdot kg^{-1}$) (Table 1). When accelerating, joint power was absorbed from foot strike until 56% of stance, followed by a large generation of power (peaking at 25.7 ± 5.4 $W \cdot kg^{-1}$, at 83% of stance) for the remainder of stance (Figure 3i). When decelerating, the ankle absorbed power from foot strike until 64% of stance and generated power for the remainder of stance (with peak power generation = 2.0 ± 3.1 $W \cdot kg^{-1}$, at 79% of stance). As a result, there was a large difference (z = 1.87, 95% CI = 1.77 to 1.96, $p < 0.001$) in net joint work between acceleration and deceleration, where net joint work was positive when accelerating (mean = 0.74 $J \cdot kg^{-1}$, 95% CI = 0.66 to 0.83 $J \cdot kg^{-1}$) but negative when decelerating (mean = -0.54 $J \cdot kg^{-1}$, 95% CI = -0.62 to -0.45 $J \cdot kg^{-1}$) (Table 1).

DISCUSSION

This study aimed to directly compare hip, knee and ankle joint mechanics when accelerating and decelerating with maximal effort. The main findings of the current study include that the net joint work displayed by the lower limbs was positive during acceleration and negative during deceleration. Additionally, acceleration was characterised by more positive joint work than deceleration for all lower limb joints (albeit not significantly from the knee), with greatest contributions from the ankle, followed by the hip and knee joints. Deceleration was characterised by more negative joint work than acceleration for all lower limb joints, with greatest contributions from the knee and hip joints (both with similar contributions) followed by the ankle joint. Deceleration was typically characterised by significantly greater peak joint moments compared to acceleration, apart from the hip flexion and ankle plantarflexion moments,

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where the opposite was true.

To our knowledge, no previous studies have compared lower limb joint mechanics during maximal effort acceleration and deceleration. However, some studies have reported similar data (albeit with notable methodological differences), hence providing some validation of our findings. For example, the joint angle, moment and power profiles (Figure 2) of the hip, knee and ankle during acceleration closely resembled that found in previous literature for sprinting with similar acceleration magnitudes (16). The average net joint work values during acceleration at the hip (0.05 to 0.48 $J \cdot kg^{-1}$), knee (0.26 to 0.56 $J \cdot kg^{-1}$) and ankle (0.79 to 1.19 $J \cdot kg^{-1}$) reported in previous work (16) also compare favourably to our estimates of 0.48, 0.37 and 0.74 $J \cdot kg^{-1}$, respectively, during acceleration. Our observation that the greatest contributors to positive joint work during acceleration were the ankle, followed by the hip and knee, is also consistent with previous work (12). To our knowledge, no previous study has quantified lower limb joint mechanics during maximal effort and unanticipated deceleration. However, one study (12) reported a rank order for contribution of the hip, knee and ankle to total positive and negative joint work performed during submaximal acceleration and deceleration (in comparison to steady state running). Williams et al. (12) found that the greatest contributions to positive joint work during acceleration came from the ankle, followed by the hip and knee joints, which aligns with the results found in the current study. However, the previous study (12) found that the greatest negative joint work during deceleration was produced by the knee (-0.73 ± 0.28 $J \cdot kg^{-1}$) and ankle (-0.55 ± 0.20 $J \cdot kg^{-1}$). This partially contrasts with the findings of the current study, which found the greatest amount of negative joint work came from the knee and hip, followed by the ankle, during deceleration (Table 1). These discrepant results may be due to protocol differences, where

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the deceleration performed in the previous study (12) was pre-planned and submaximal, whereas the present study tested rapid and unanticipated deceleration to more closely replicate demands required in sporting settings (1). The greater decelerative demands of our protocol are supported by the greater total negative joint work for all joints compared to that previously reported (12). Subsequently, our findings (i.e., that the knee and hip produce the greatest amount of negative work during deceleration) may reflect an alternative mechanical strategy when decelerating with maximal effort, compared to the submaximal strategy employed in previous work (12).

A net extension moment at the hip was present during the majority of stance for both acceleration and deceleration in this study, suggesting that the hip extensors (gluteus maximus, hamstrings and adductor magnus) are critical to performance of such tasks (Figure 3e). The hip extension moment was associated with positive net joint work during acceleration (Figure 3d, Figure 4a), suggesting that the hip extensors are contracting concentrically. In comparison, the hip extension moment during deceleration was associated with negative net joint work (Figure 3d, Figure 4a), suggesting that the hip extensors were being loaded eccentrically. Performing rapid, high load, eccentric contractions can increase the risk of musculotendinous injury, where the muscles of the hamstrings and groin are frequently injured in sports requiring the performance of rapid acceleration and deceleration (4, 29). Whilst the hamstrings may be susceptible to injury during the swing phase of sprinting (30), we observed that when decelerating the early stance phase was characterised by a relatively flexed hip and extended knee (suggesting stretching of the hamstrings), whilst there was also a simultaneous hip extension and knee flexion moment, likely placing high loads on the hamstrings and potentially making them vulnerable to injury (30). Additionally, during late stance when accelerating and

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decelerating, we observed negative (eccentric) work with a simultaneous hip flexion moment, suggesting the hip flexors (e.g., adductor longus, iliopsoas etc.) may be vulnerable to injury during this period. These conclusions are supported by observations that accelerations and decelerations are common injury mechanisms for the hamstrings (4) and groin (29). This may have implications for the importance of eccentric hamstring and hip flexor/adductor strength for the performance of accelerations and decelerations, as well as mitigating the risk of injury.

Our findings indicate that there is predominately a net extension moment produced at the knee for both acceleration and deceleration conditions, suggesting that the mechanics of the knee extensors (such as the quadriceps) are important for the performance of both tasks (Figure 3e). This knee extensor work is predominately positive when accelerating (Figure 3e, Figure 4b), suggesting the knee extensors are acting concentrically. In contrast, deceleration was primarily characterised by negative net joint work by the knee extensors, suggesting that the knee extensors are subjected to large eccentric demands (Figure 3e, Figure 4b). This is supported by the previously established relationship between eccentric quadricep strength and the production of horizontal braking forces (31), and modelling studies showing the direct contribution of the quadriceps to braking forces during a variety of high intensity weightbearing movement patterns (21, 32, 33). Training programs focusing on knee extensor strength (particularly eccentrically) could contribute to the improvement of deceleration performance. However, force production from the quadriceps is associated with increased loads on the anterior cruciate ligament (ACL) via its contribution to anterior shear forces at the tibia when knee flexion angles are small (34), which may partly explain why ACL injuries often occur during rapid decelerations (3, 6, 7). Given the previously described importance of the quadriceps during deceleration, optimising the

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function of the quadriceps together with the muscles that produce posterior shear forces at the tibia (thus unloading the ACL e.g., hamstrings and soleus (34)) may be important for achieving enhanced performance whilst at the same time minimising anterior shear forces at the knee.

Both acceleration and deceleration were associated with substantial demands on the ankle plantar flexors (Figure 3f, Figure 4c). This is unsurprising as the plantar flexors have been shown to contribute to body mass support and forward propulsion in a range of tasks, such as walking (14) accelerative running (33) and sidestep cutting (21). Such high demands on the ankle plantar flexors may partly explain why calf injuries are common in sports that require rapid accelerations and decelerations (5). Acceleration and deceleration, however, were also observed to have notable differences in ankle mechanics, which may have implications for performance and injury risk. For example, acceleration was characterised by a dorsi-flexed ankle at initial contact, which continued dorsiflexing until 55% of stance, at which point the ankle rapidly plantarflexed (Figure 3c). In contrast, the ankle was plantarflexed at initial contact of deceleration, and the plantarflexion angle increased for the first 8% of stance before following a similar pattern to the accelerative condition (i.e., increased dorsi-flexion followed by plantar flexion) (Figure 3c). This kinematic pattern suggests our participants utilised a heel-strike pattern when decelerating, and a forefoot strike pattern when accelerating (Figure 2a and Figure 2b). This is supported by the net dorsi-flexion moment (whilst absorbing power, Figures 3f and 3i) for the first 8% of stance in the deceleration condition, which was not observed in the acceleration condition. Thus, eccentric training of the ankle-dorsiflexors may be an important component of interventions aiming to support deceleration performance (35). Additionally, these findings may have implications for ACL injuries, which are known to occur during early foot-strike of deceleration manoeuvres (3,

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6, 7). During the early foot-strike period, the heel strike pattern (and consequent dorsi-flexion moment) during deceleration suggests limited recruitment of the ankle plantar flexors such as the soleus and gastrocnemius, which are known to induce posterior and anterior shear force at the knee, respectively (34). Consequently, interventions aiming to reduce ACL injury risk during deceleration by targeting the ankle plantar flexors may have limited benefit, unless participants can adopt or naturally have a mid or forefoot strike pattern. It should be noted that exploring injury risk is beyond the scope of this investigation (owing to its cross-sectional design), however, further research could investigate the effects of gait retraining in deceleration to reduce injuries associated with using a heel strike pattern.

This study has several limitations that should be considered when interpreting the findings. Firstly, when using skin-mounted reflective markers to estimate joint angles (and subsequent joint kinetics), soft tissue artefact error can occur, meaning the location of skin-mounted markers may not always perfectly represent the underlying anatomical bony landmark/segments (36). However, we used a global optimisation approach to compute joint angles, which has been shown to minimise the influence of soft tissue artefact compared to other methods (37). Secondly, this study aimed to mimic the random and reactive nature of decelerations that occur in field sports settings (1) (via the use of timing gates), but we acknowledge that the laboratory setting where the experimental protocol was conducted is not an exact replication of what would occur on the field. Timing gates and a laboratory setting were chosen to control extraneous factors and support the internal validity of the study. Despite this, the magnitude of the average forward velocity of the COM at foot-strike differed by 5% between conditions. However, we verified that our results were not affected by this difference (see

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Supplemental Digital Content, http://links.lww.com/MSS/C997). Finally, caution is advised when generalising our findings to an elite athletic or previously injured population, as sprinting gait patterns may differ between these groups and the healthy, recreationally active adult participants in the current study. The use of non-elite athletes may also partly explain the heterogeneous execution of maximal decelerations across participants (note the larger SDs in the deceleration condition in Figure 3, see Supplemental Figures 3 and 4, Supplemental Digital Content, for individual trial variability, http://links.lww.com/MSS/C997), as it was performed as a rapid reaction to an external stimulus without any prior coaching from researchers. Due to the sample size, participants were not able to be sub-grouped into specific deceleration strategies/techniques, nor were we able to test for sex-specific differences. Further work with larger sample sizes may be able to elucidate differing deceleration strategies, including variations that may occur in elite athletic or previously injured populations, potentially incorporating field settings.

CONCLUSIONS

In summary, this study investigated the mechanics of the hip, knee and ankle joints during the performance of maximal effort acceleration and rapid, unanticipated deceleration. It was found that the net joint work was positive during acceleration, with the dominant contributors being the hip extensors during early stance and the ankle plantar flexors during late stance. Net joint work during deceleration was negative, where the knee and hip extensors produced the greatest contributions to negative joint work. Peak joint moments were typically greater during deceleration compared to acceleration, apart from the hip flexion and ankle plantar flexion moments where the opposite was true. These findings can inform training interventions

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that aim to enhance performance of acceleration and deceleration tasks and mitigate the associated risk of injury.

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Acknowledgments

This study was partly funded from the Faculty of Health Sciences Funding Support for Honours and Masters by Coursework with Minor Thesis scheme, provided by the Australian Catholic University. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

Conflicts of interest

A/Prof David Opar is listed as a co-inventor on a patent, filed by the Queensland University of Technology (QUT), for a field-testing device of eccentric hamstring strength, which is now known commercially as the NordBord. A/Prof Opar has received revenue distributions from QUT based on revenue that QUT has generated through the commercialisation of his intellectual property.

A/Prof Opar is a minority shareholder in Vald Performance Pty Ltd, the company responsible for commercialization of the NordBord and the SmartSpeed timing gates, the latter which were utilised in the current manuscript. A/Prof Opar has received research funding from Vald Performance, for work unrelated to the current manuscript. A/Prof Opar was previously the Chair of the Vald Performance Research Committee, a role that was unpaid.

A/Prof Opar has family members who are minor shareholders and/or employees of Vald Performance.

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FIGURE LEGENDS

Figure 1. Illustration of the sequence from biomechanical modelling pipeline. Sequence steps are a) generic biomechanical model with joint axes and degrees of freedom displayed for hip (adduction, internal rotation and flexion about the x, y and z axes), knee (flexion about the z axis), ankle (dorsiflexion about the z axis) and subtalar (inversion about the x axis), b) scaling of generic model to individual's arthrometry and marker registration, c) inverse kinematics to calculate joint angles (example shown at the knee), and d) inverse dynamics to calculate net moments acting at a joint. Visualisation created in OpenSim version 4.2 (23, 24).

Figure 2. Illustration of the average kinematics and ground reaction force (GRF) vector (black arrow) for twenty participants during acceleration (a) and deceleration (b), and the mean (solid line) and SD (shaded area) of bodyweight (BW) normalised GRFs in the anteroposterior (c), vertical (d) and mediolateral (e) directions across stance. Positive values indicate anterior, vertical and lateral directed GRFs for panels c, d and e, respectively.

Figure 3. Mean (line) and SD (shaded area) joint angles (°, first row), joint moments ($N \cdot m \cdot kg^{-1}$, second row) and joint powers ($W \cdot kg^{-1}$, third row) for the hip (first column), knee (second column), and ankle (third column) joints during the stance phase of acceleration (grey) and deceleration (blue). For joint angles and moments, positive values indicate flexion (hip and knee), and dorsiflexion (ankle). For joint power, positive values indicate generation, and negative values indicate absorption.

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Figure 4. Net joint work ($J \cdot kg^{-1}$) produced at the hip (a), knee (b), and ankle (c) joints in relation to the net centre-of-mass acceleration magnitude ($m \cdot s^{-2}$) for acceleration (grey) and deceleration (blue) conditions. Density graphs on the top and right side for each joint plot indicate the net joint work ($J \cdot kg^{-1}$) and acceleration ($m \cdot s^{-2}$) distribution, respectively. For the x-axis (net acceleration magnitude ($m \cdot s^{-2}$)) positive values indicate forward acceleration and negative values indicate deceleration

SUPPLEMENTAL DIGITAL CONTENT

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Figure 1

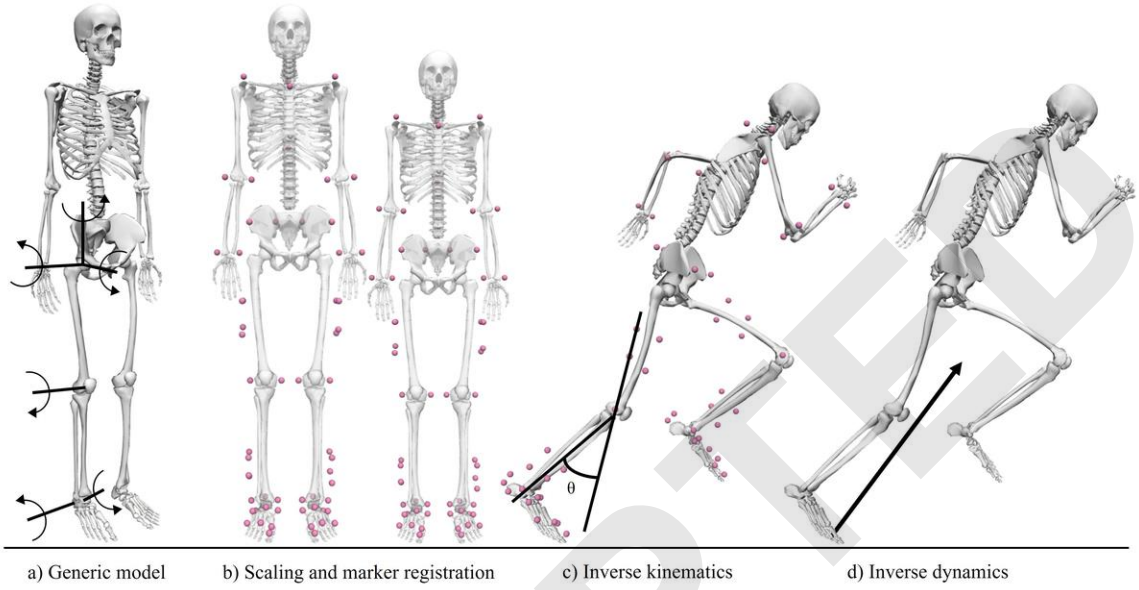

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Figure 2

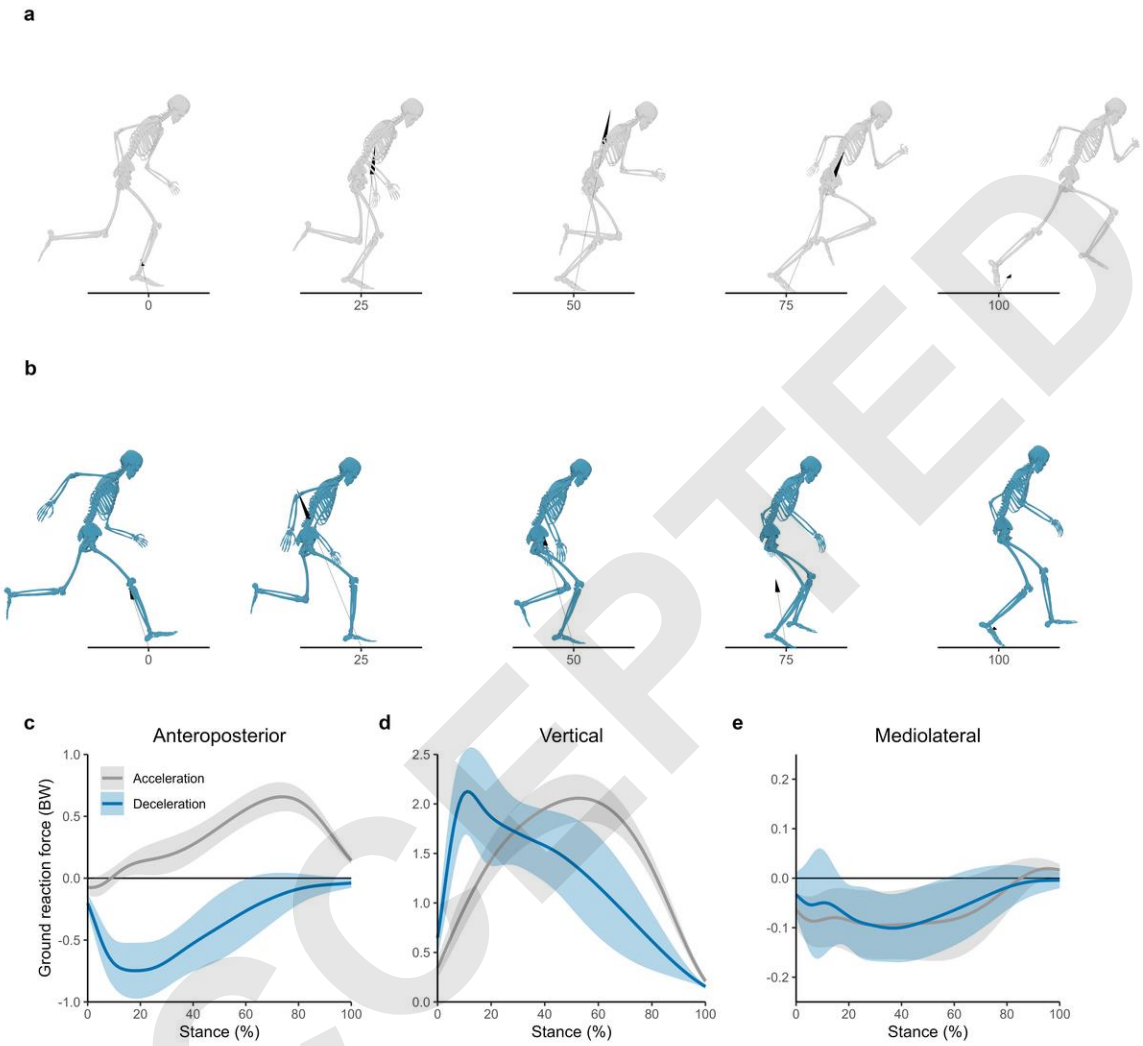

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Figure 3

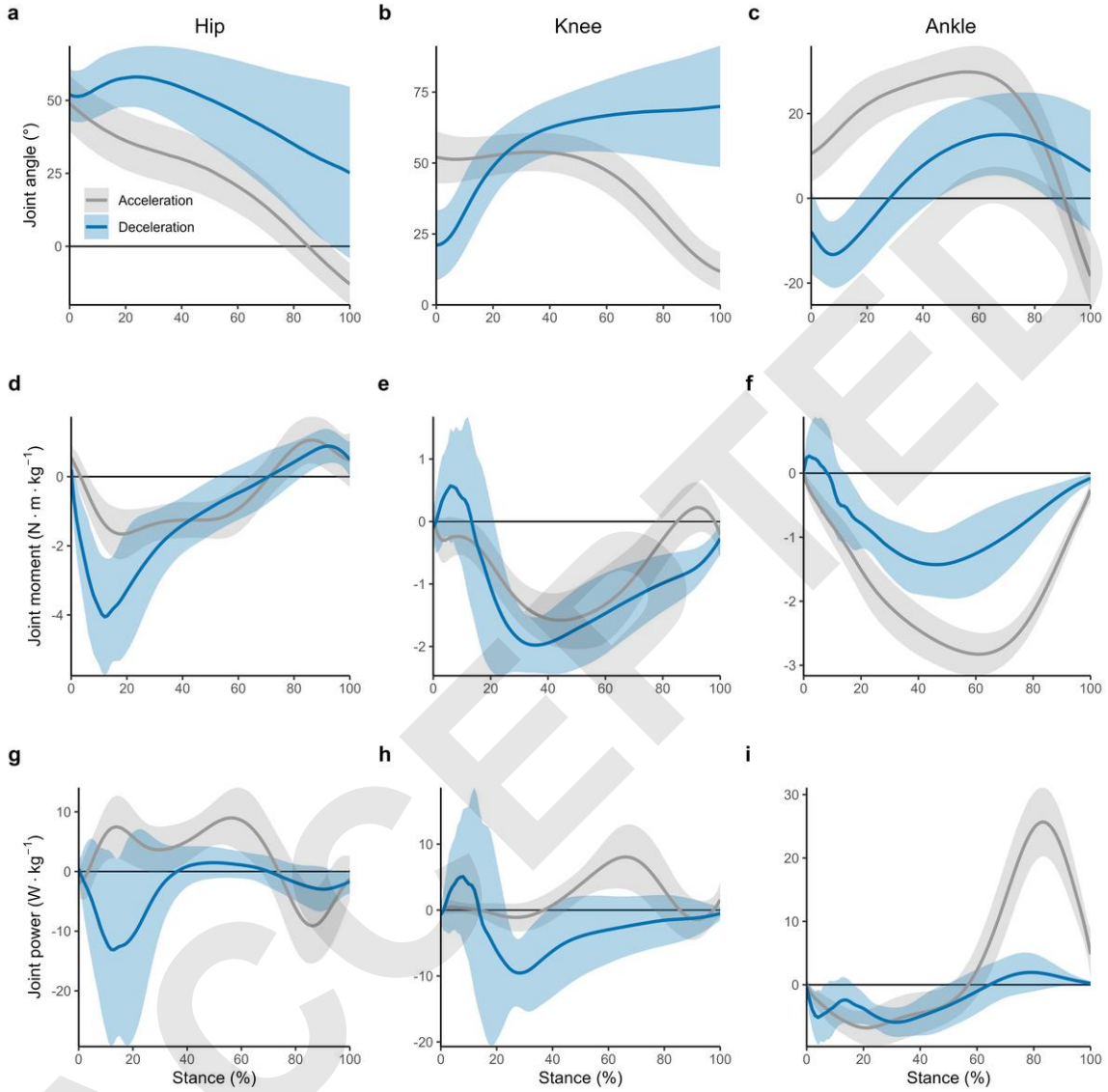

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Figure 4

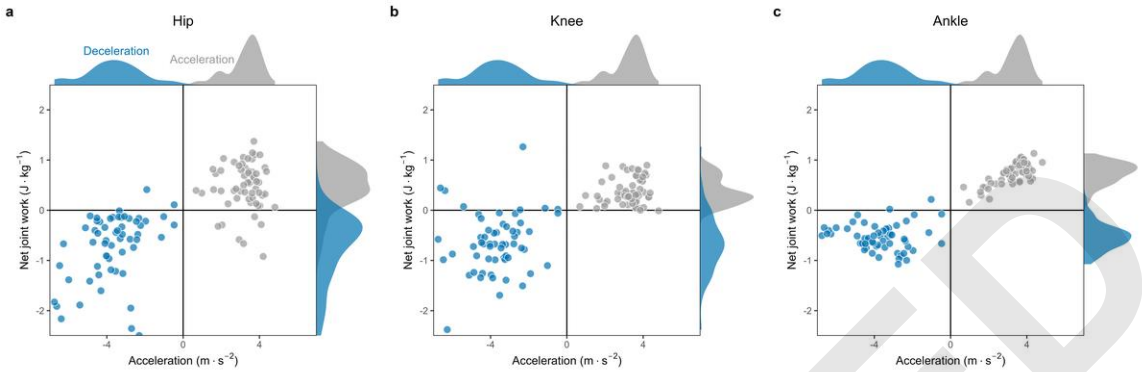

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Table 1. Marginal mean and 95% confidence interval for joint moment peaks ($N \cdot m \cdot kg^{-1}$), net joint work ($J \cdot kg^{-1}$) and total positive and negative joint work ($J \cdot kg^{-1}$) for the hip, knee and ankle joints during acceleration and deceleration conditions. Positive joint moment values indicate flexion (hip and knee) and dorsiflexion (ankle).

Joint	Acceleration	Deceleration	z	p
	Mean (lower to upper)	Mean (lower to upper)	Estimate (lower to upper)	
Peak joint moment ($N \cdot m \cdot kg^{-1}$)				
Hip flexion	1.43 (1.25 to 1.62)	1.02 (0.83 to 1.20)	0.72 (0.43 to 1.00)	<0.001
Knee flexion	0.41 (0.13 to 0.69)	1.33 (1.05 to 1.61)	-0.96 (-1.23 to -0.69)	<0.001
Ankle dorsiflexion	0.02 (-0.08 to 0.11)	0.65 (0.56 to 0.75)	-1.50 (-1.70 to -1.30)	<0.001
Hip extension	-1.88 (-2.38 to -1.39)	-4.97 (-5.46 to -4.48)	1.50 (1.31 to 1.68)	<0.001
Knee extension	-1.66 (-1.85 to -1.47)	-2.30 (-2.49 to -2.10)	1.07 (0.85 to 1.29)	<0.001
Ankle plantarflexion	-2.86 (-3.01 to -2.70)	-1.76 (-1.91 to -1.60)	-1.61 (-1.76 to -1.46)	<0.001
Net joint work ($J \cdot kg^{-1}$)				
Hip	0.48 (0.29 to 0.66)	-0.73 (-0.91 to -0.55)	1.46 (1.24 to 1.68)	<0.001
Knee	0.37 (0.24 to 0.50)	-0.65 (-0.78 to -0.52)	1.50 (1.28 to 1.73)	<0.001
Ankle	0.74 (0.66 to 0.83)	-0.54 (-0.62 to -0.45)	1.87 (1.77 to 1.96)	<0.001
Positive joint work ($J \cdot kg^{-1}$)				
Hip	0.82 (0.74 to 0.90)	0.24 (0.16 to 0.32)	1.56 (1.36 to 1.76)	<0.001
Knee	0.49 (0.39 to 0.58)	0.37 (0.27 to 0.47)	0.40 (0.08 to 0.71)	0.014
Ankle	1.24 (1.18 to 1.29)	0.14 (0.08 to 0.20)	1.91 (1.83 to 1.99)	<0.001
Negative joint work ($J \cdot kg^{-1}$)				
Hip	-0.35 (-0.53 to -0.17)	-0.97 (-1.15 to -0.79)	1.03 (0.77 to 1.29)	<0.001
Knee	-0.12 (-0.20 to -0.03)	-1.01 (-1.10 to -0.93)	1.67 (1.48 to 1.86)	<0.001
Ankle	-0.49 (-0.55 to -0.43)	-0.67 (-0.74 to -0.61)	0.93 (0.67 to 1.19)	<0.001

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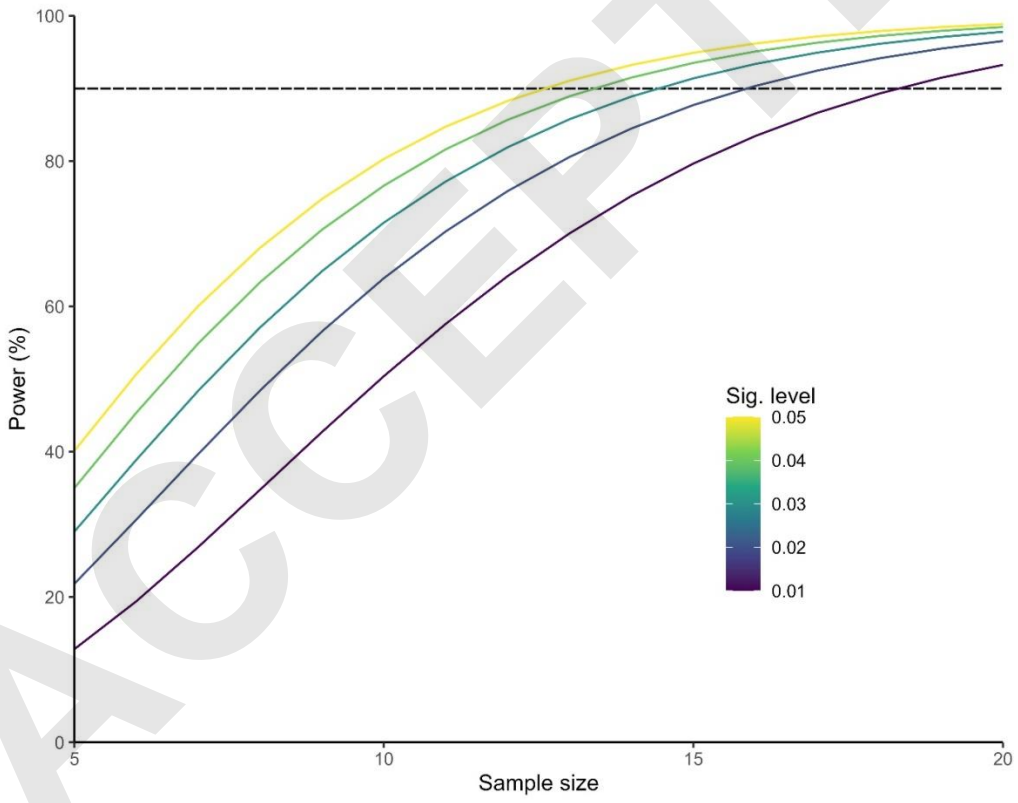
Supplemental Digital Content 1

This document provides a visualisation of our power analysis calculations. As described in the main article, we adopted conservative input parameters (d = 1, α = .01 and power =.90) for our sample size calculation, despite previous work suggesting higher effect sizes (i.e., >1) were anticipated. This is because 1) our analysis included more variables than just joint work (e.g., peak joint moments) which were not reported by previous work, and 2) the study used as a basis for sample size calculations had a small sample size (ten participants), thus their effect sizes estimates may be inflated due to lowered statistical power.



Supplementary Figure 1. The influence of sample size on statistical power (for paired t-test comparisons) at five different significance (alpha) levels. Note that an anticipated effect size of 1 was used for all calculations, with the desired power (90%) indicated by the dashed horizontal line.

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Supplemental Digital Content 2

This document provides a visualisation non-sagittal lower limb joint mechanics (i.e., for the hip in the frontal and transverse planes, and the subtalar joint). Note that since joint power is scalar, hip power was computed as the sum of its three orthogonal components. Subsequently, Supplementary Figure 2 (below) does not show hip power (as it is reported in the main results).

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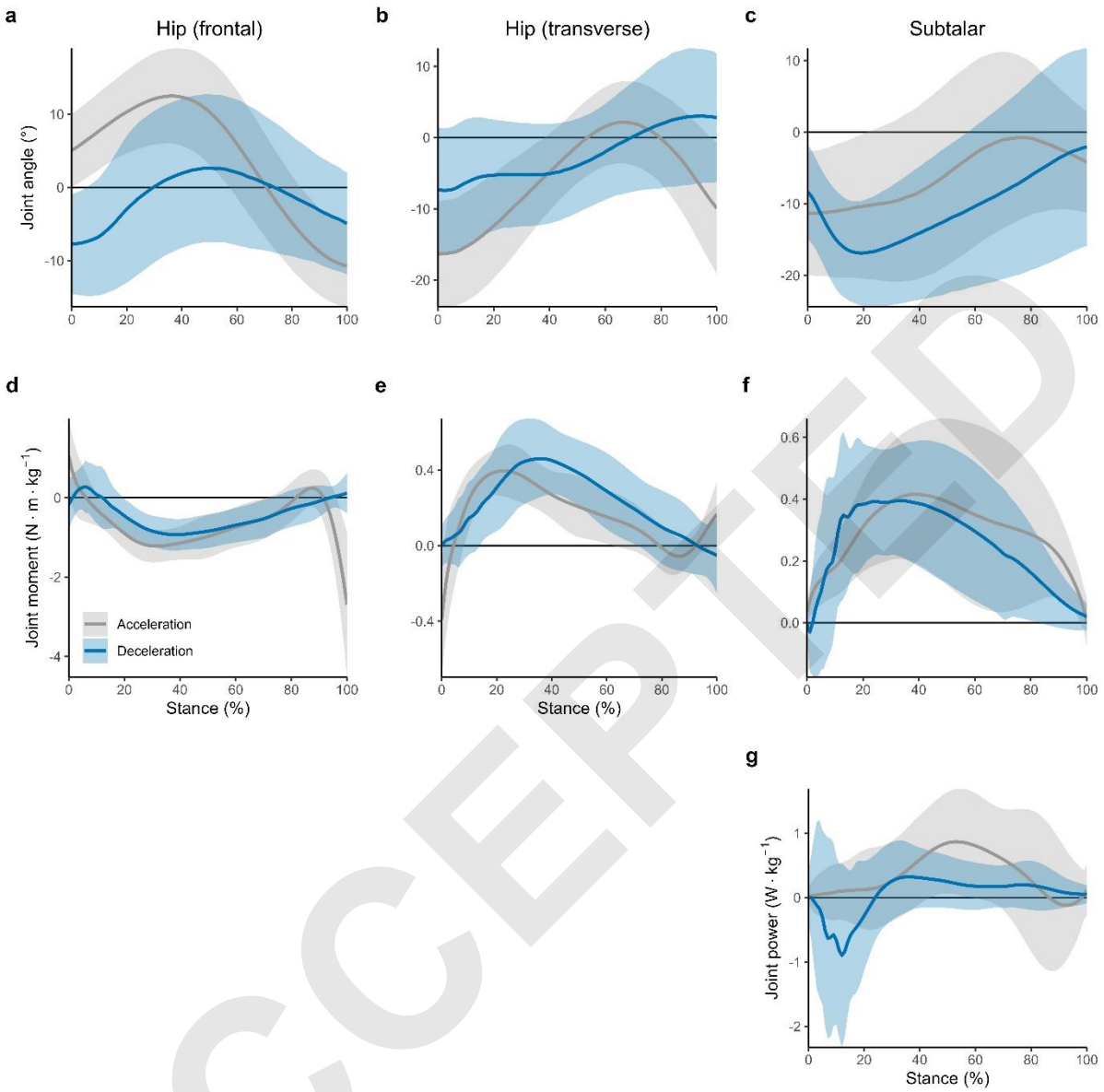

Supplementary Figure 2. Mean (line) and SD (shaded area) joint angles (°, first row), joint moments ($N{\cdot}m{\cdot}kg^{-1}$, second row) and joint powers ($W{\cdot}kg^{-1}$, third row) for the hip in the frontal (first column) and transverse (second column) planes, and the subtalar (third column) joints during the stance phase of acceleration (grey) and deceleration (blue). For joint angles and moments, positive values indicate adduction, internal rotation and inversion for the three columns, respectively. For joint power, positive values indicate generation, and negative values indicate absorption.

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Supplemental Digital Content 3

This document provides our statistical results the include the forward centre-of-mass velocity at foot-strike as a covariate, performed to ensure our conclusions were robust. The statistical model used in the main manuscript took the form:

$$\begin{aligned} \text{KineticVar}_i &\sim N\left(\alpha_{j[i]} + \beta_1(\text{Task}_{\text{decel}}), \sigma^2\right) \\ \alpha_j &\sim N\left(\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right), \text{ for Participant j = 1, ... ,J} \end{aligned}$$

Thus each kinetic variable of interest (KineticVar_i) was modelled using task as a fixed effect, and participant as a random intercept. This original model did not include forward centre-of-mass velocity at foot-strike as a covariate, owing to its complex inter-relationship with the experimental condition (i.e., acceleration and deceleration, which represents change in velocity). However, it is possible that differing velocities at foot-strike may complicate the interpretation of our results, thus this document provides the results of an alternate statistical model:

$$\begin{aligned} \text{KineticVar}_i &\sim N\left(\alpha_{j[i]} + \beta_1(\text{Task}_{\text{decel}}) + \beta_2(COM_{vel}), \sigma^2\right) \\ \alpha_j &\sim N\left(\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right), \text{ for Participant j = 1, ... ,J} \end{aligned}$$

This updated model includes forward centre-of-mass velocity at foot-strike (COM_{vel}) as a covariate, thus the subsequently reported results are adjusted for this covariate i.e., the marginal means and effect sizes are conditional on the forward centre-of-mass velocity at foot-strike = 3.81 $m \cdot s^{-1}$ (mean across all trials). As shown in Supplementary Table 1 (below), our numbers differ only slightly from that reported in the main manuscript, and all our conclusions (i.e., significant and non-significant results) are identical.

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Supplementary Table 1. Marginal mean and 95% confidence interval for joint moment peaks ($N \cdot m \cdot kg^{-1}$), net joint work ($J \cdot kg^{-1}$) and total positive and negative joint work ($J \cdot kg^{-1}$) for the hip, knee and ankle joints during acceleration and deceleration conditions. Note that all estimates were adjusted for the forward centre-of-mass velocity at foot-strike. Positive joint moment values indicate flexion (hip and knee) and dorsiflexion (ankle).

Joint	Acceleration	Deceleration	z	p
	Mean (lower to upper)	Mean (lower to upper)	Estimate (lower to upper)	
Peak joint moment ($N \cdot m \cdot kg^{-1}$)				
Hip flexion	1.42 (1.24 to 1.59)	1.04 (0.86 to 1.21)	0.65 (0.34 to 0.96)	<0.001
Knee flexion	0.37 (0.10 to 0.65)	1.37 (1.09 to 1.65)	-1.04 (-1.32 to -0.75)	<0.001
Ankle dorsiflexion	0.02 (-0.08 to 0.12)	0.65 (0.55 to 0.75)	-1.49 (-1.70 to -1.27)	<0.001
Hip extension	-1.78 (-2.24 to -1.32)	-5.08 (-5.55 to -4.62)	1.60 (1.41 to 1.80)	<0.001
Knee extension	-1.68 (-1.89 to -1.46)	-2.28 (-2.49 to -2.07)	1.02 (0.78 to 1.26)	<0.001
Ankle plantarflexion	-2.82 (-2.95 to -2.70)	-1.80 (-1.92 to -1.67)	-1.50 (-1.66 to -1.34)	<0.001
Net joint work ($J \cdot kg^{-1}$)				
Hip	0.47 (0.27 to 0.66)	-0.72 (-0.92 to -0.52)	1.43 (1.20 to 1.66)	<0.001
Knee	0.37 (0.24 to 0.51)	-0.66 (-0.79 to -0.52)	1.51 (1.28 to 1.75)	<0.001
Ankle	0.73 (0.65 to 0.82)	-0.52 (-0.61 to -0.44)	1.84 (1.73 to 1.94)	<0.001
Positive joint work ($J \cdot kg^{-1}$)				
Hip	0.81 (0.72 to 0.89)	0.26 (0.18 to 0.34)	1.47 (1.27 to 1.67)	<0.001
Knee	0.48 (0.38 to 0.57)	0.38 (0.28 to 0.47)	0.33 (0.00 to 0.67)	0.050
Ankle	1.22 (1.17 to 1.27)	0.16 (0.11 to 0.20)	1.86 (1.78 to 1.95)	<0.001
Negative joint work ($J \cdot kg^{-1}$)				
Hip	-0.34 (-0.53 to -0.16)	-0.97 (-1.16 to -0.79)	1.04 (0.76 to 1.32)	<0.001
Knee	-0.10 (-0.18 to -0.02)	-1.02 (-1.11 to -0.94)	1.72 (1.52 to 1.91)	<0.001
Ankle	-0.49 (-0.55 to -0.42)	-0.68 (-0.74 to -0.61)	0.94 (0.66 to 1.22)	<0.001

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Supplemental Digital Content 4

This document provides a visualisation of individual variability observed for ground reaction forces (Supplementary Figure 3) and lower limb joint mechanics (Supplementary Figure 4).

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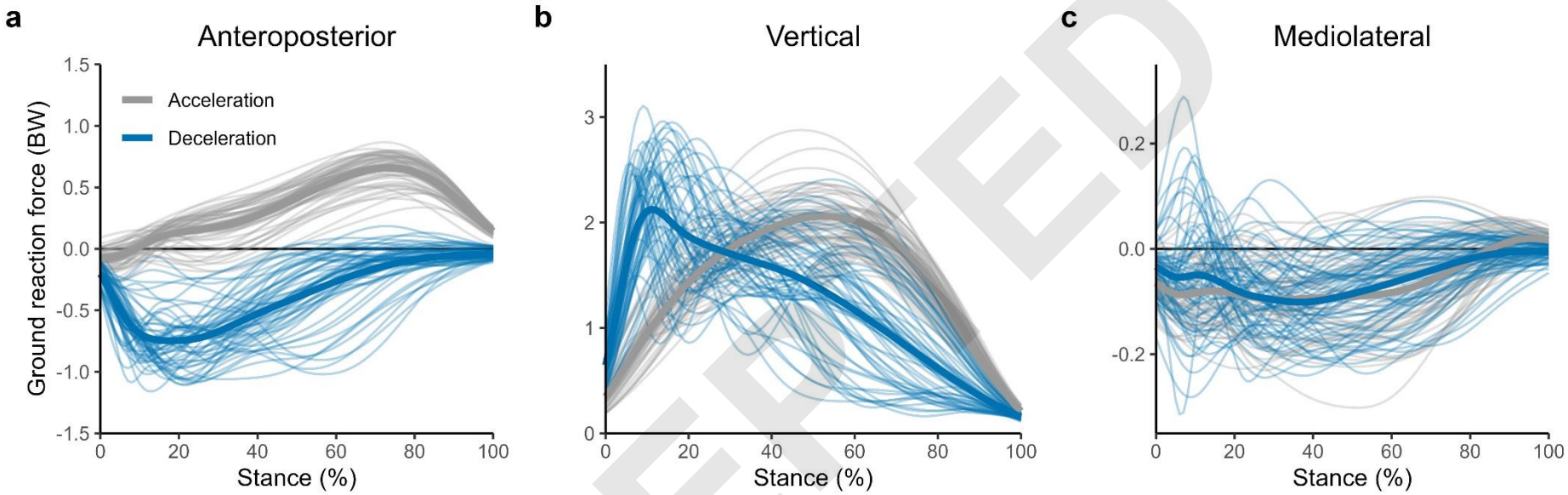

Supplementary Figure 3. Illustration of the average (thick lines) and individual-trial (thin lines) ground reaction forces (GRFs) for twenty participants during acceleration (grey) and deceleration (blue) in the anteroposterior (a), vertical (b) and mediolateral (c) directions across stance. Positive values indicate anterior, vertical and lateral directed GRFs for panels a, b and c, respectively

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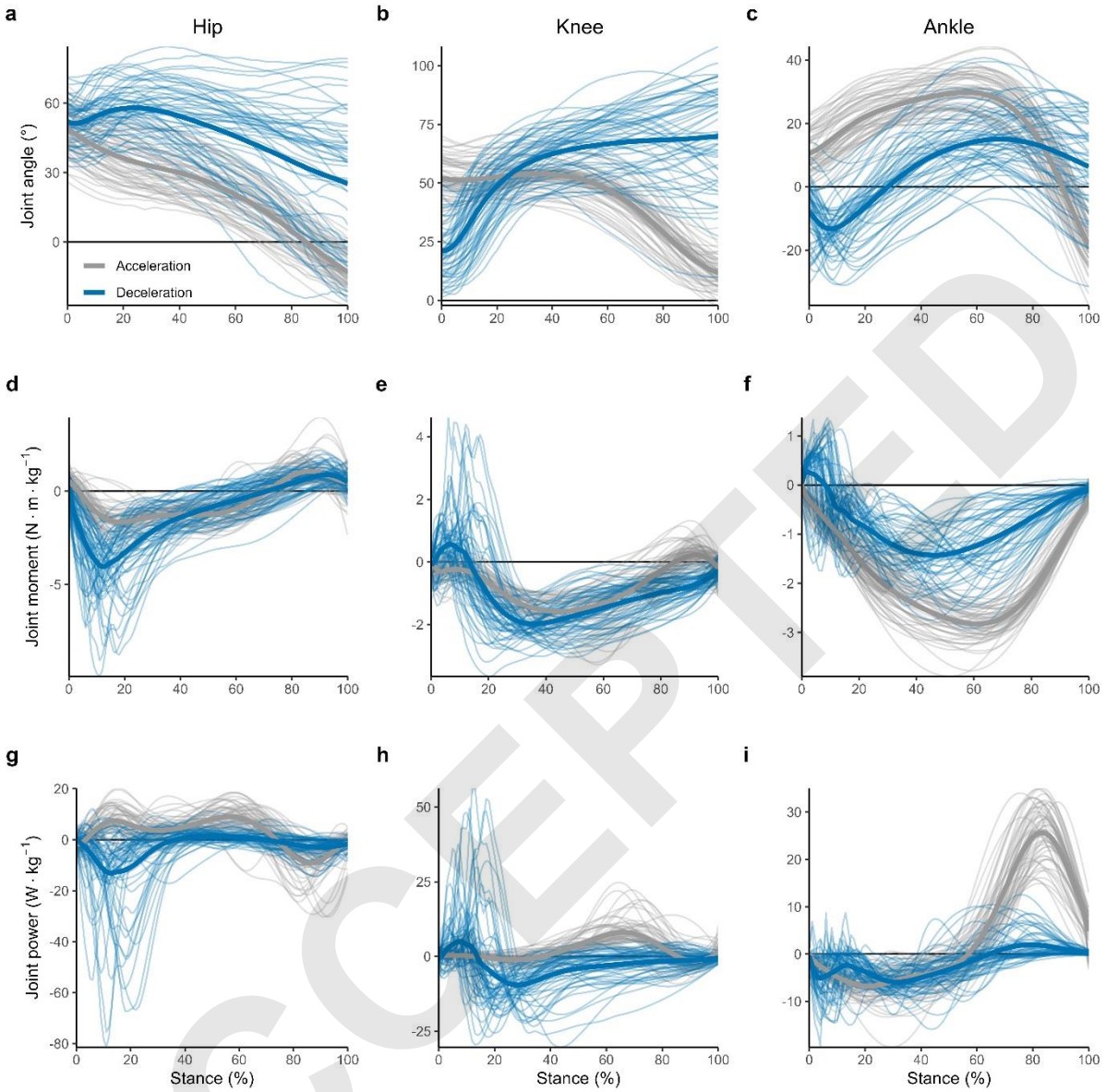

Supplementary Figure 4. Mean (thick line) and individual-trial (thin line) joint angles (°, first row), joint moments (N·m·kg^{-1}, second row) and joint powers (W·kg^{-1}, third row) for the hip (first column), knee (second column), and ankle (third column) joints during the stance phase of acceleration (grey) and deceleration (blue). For joint angles and moments, positive values indicate flexion (hip and knee) and dorsiflexion (ankle). For joint power, positive values indicate generation, and negative values indicate absorption.

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Supplemental Digital Content 5

This document provides example setup files for inverse kinematics, inverse dynamics, and kinematics analysis in OpenSim version 4.2.

Inverse kinematics


```
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<OpenSimDocument Version="40000">
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    <constraint_weight>Inf</constraint_weight>
    <accuracy>1e-005</accuracy>
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          <weight>1</weight>
        </IKMarkerTask>
        <IKMarkerTask name="L.Acromion">
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</IKMarkerTask>
<IKMarkerTask name="R.Thigh.Post.Sup">
```


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  <IKMarkerTask name="R.Thigh.Ant.Sup">
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  </IKMarkerTask>
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  <IKMarkerTask name="R.Shank.Post.Sup">
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  <IKMarkerTask name="R.Ankle.Lat">
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</IKMarkerTask>
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</IKMarkerTask>
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</IKMarkerTask>
```


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</IKMarkerTask>
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    </objects>
    <groups/>
  </IKTaskSet>
  <marker_file>Marker.trc</marker_file>
  <coordinate_file>Unassigned</coordinate_file>
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  <report_errors>true</report_errors>
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  <report_marker_locations>true</report_marker_locations>
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</OpenSimDocument>
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Inverse dynamics


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  <input_directory/>
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  <external_loads_file>External_loads.xml</external_loads_file>
  <coordinates_file>IK.mot</coordinates_file>
 <lowpass_cutoff_frequency_for_coordinates>15</lowpass_cutoff_frequency_for_coordinates>
  <output_gen_force_file>inverse_dynamics.sto</output_gen_force_file>
 </InverseDynamicsTool>
</OpenSimDocument>
```


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Kinematic analysis


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    <initial_time>0</initial_time>
    <final_time>1</final_time>
<solve_for_equilibrium_for_auxiliary_states>false</solve_for_equilibrium_for_auxiliary_states
>
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    <maximum_integrator_step_size>1</maximum_integrator_step_size>
    <minimum_integrator_step_size>1e-05</minimum_integrator_step_size>
    <integrator_error_tolerance>1e-05</integrator_error_tolerance>
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        <Kinematics name="Kinematics">
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          <end_time>1</end_time>
          <step_interval>1</step_interval>
          <in_degrees>true</in_degrees>
        </Kinematics>
      </objects>
```


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    </AnalysisSet>
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    <coordinates_file>IK.mot</coordinates_file>
  <lowpass_cutoff_frequency_for_coordinates>15</lowpass_cutoff_frequency_for_coordinates>
  </AnalyzeTool>
</OpenSimDocument>
```

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